Thermodynamics

Manav M Nair

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1 Introduction

1.1 State Variables

These are variables which we can define for a thermodynamic system that describes the system at any point in time. The state variables for a system may change if we do different processes on it but if at a later time, the system were to be brought back to how it was before, then the state variables must also return to the same value as before.

The two common state state variables that we can measure with our senses or using instruments are Pressure(P) and Volume(V). In the absence of any constraints, these two state variables can change independently of each other. That is, a gas can take any possible value of pressure at a particular value of volume, and vice versa. But when certain constraints are introduced, they lock the variables into a single function such that if I vary one, the other one also changes. We shall see one such constraint in the next section.

1.2 Quasi-Static process

It is a process that is so slow that the system is assumed to be close to (or at) equilibrium at each instant of time.

Need for a Quasi-static process: Equations of state that connect the different state variables together are only described when the system is in a state of equilibrium.

If processes happen at time-scales that are at least an order of magnitude larger than the relaxation times of the system, then the process can be assumed to be quasi-static.

All quasi-static processes are not reversible but all reversible process must be quasi-static.

2 The Zero'th Law of Thermodynamics

2.1 Thermal Equilibrium

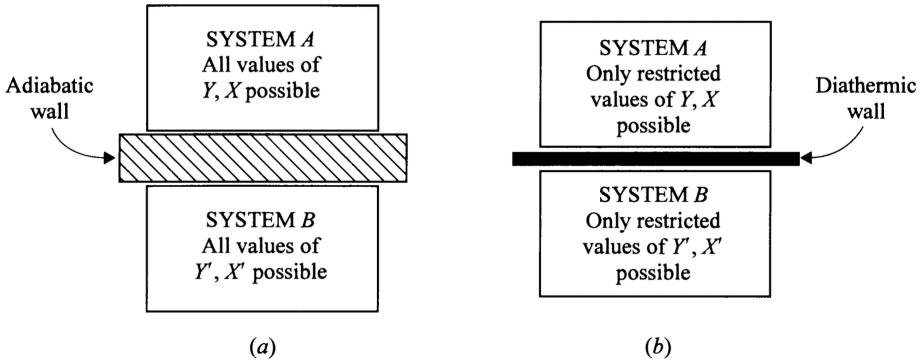
In thermodynamics, an isolated system being in thermal equilibrium is introducing a constraint so as to make its state variables no longer independent of each other. That means I can write one of the variables as a function of another variables and there exist a function $f(P, V)$ such that

$$f(P, V) = 0$$

If I have two systems in thermal equilibrium but separated by an adiabatic wall, both the systems can independently be in thermal equilibrium. That is, $f(P_1, V_1) = f(P_2, V_2) = 0$. In

other words, an equilibrium state of the first system can coexist with any equilibrium state of the second system.

But when the two systems are separated by a diathermic wall instead, and they attain thermal equilibrium, the state variables of the combined system will change spontaneously until an equilibrium state of the combined system has be attained. Since the four state variables are then related, I can write $f(P_1, V_1, P_2, V_2) = 0$



Thermodynamic Equilibrium A system which is simultaneously in mechanical, chemical and thermal equilibrium is said to be in thermodynamic equilibrium. There exists equations of state that govern a thermodynamics equilibrium.

2.2 The Zeroth Law

Definition: Say I have two systems — $\mathcal{A}$ & $\mathcal{B}$, separated by an adiabatic wall and are separately in thermal equilibrium with a third system $\mathcal{C}$. Then it is found that $\mathcal{A}$ and $\mathcal{B}$ are in thermal equilibrium with each other.

Say I'm able to define the state of $\mathcal{A}$, $\mathcal{B}$ and $\mathcal{C}$ by state variables P and V. Since $\mathcal{A}$ and $\mathcal{C}$ are in thermal equilibrium and are separated by a **diathermic wall**, the state variables of the combined system are not independent and there exist a function F_1 such that

$$F_1(P_\mathcal{A}, V_\mathcal{A}, P_\mathcal{C}, V_\mathcal{C}) = 0 \tag{2.1}$$

$$\implies P_\mathcal{C} = f_1(P_\mathcal{A}, V_\mathcal{A}, V_\mathcal{C}) \tag{2.2}$$

and similarly for $\mathcal{B}$ and $\mathcal{C}$,

$$F_2(P_\mathcal{B}, V_\mathcal{B}, P_\mathcal{C}, V_\mathcal{C}) = 0 \tag{2.3}$$

$$\implies P_\mathcal{C} = f_2(P_\mathcal{B}, V_\mathcal{B}, V_\mathcal{C}) \tag{2.4}$$

Similarly, since $\mathcal{A}$ is in thermal equilibrium with $\mathcal{B}$ according to zeroth law, There exist a function F_3 such that

$$F_3(P_\mathcal{A}, V_\mathcal{A}, P_\mathcal{B}, V_\mathcal{B}) = 0 \tag{2.5}$$

We see that equation (2.2) and (2.4) has V_c but (2.5) does not. That makes sense because by the zeroth law, the equilibrium between $\mathcal{A}$ and $\mathcal{B}$ does not depend on $\mathcal{C}$. Thus, if I were to equate (2.2) and (2.4) (since they both equal $P_{\mathcal{C}}$), I need to write it in a form where $V_{\mathcal{C}}$ cancels from both sides. For this, I shall write f_1 and f_2 as

$$f_1 = \Phi(P_{\mathcal{A}}, V_{\mathcal{A}}) \cdot \zeta(V_{\mathcal{C}}) + \eta(V_{\mathcal{C}})$$
$$f_2 = \Phi(P_{\mathcal{B}}, V_{\mathcal{B}}) \cdot \zeta(V_{\mathcal{C}}) + \eta(V_{\mathcal{C}})$$

Since f_1 and f_2 are equal, I can write

$$\Phi(P_{\mathcal{A}}, V_{\mathcal{A}}) = \Phi(P_{\mathcal{B}}, V_{\mathcal{B}})$$

Note that $\Phi(P_{\mathcal{A}}, V_{\mathcal{A}}) - \Phi(P_{\mathcal{B}}, V_{\mathcal{B}}) = 0$ is simply our function $F_3(P_{\mathcal{A}}, V_{\mathcal{A}}, P_{\mathcal{B}}, V_{\mathcal{B}})$ according to (2.5)

If we call the scalar value of this function as "*empirical temperature*", we see that any two systems at thermal equilibrium with each other have the same value of this quantity called temperature. Thus, I can write the same for system $\mathcal{C}$ too.

$$\boxed{\Phi(P_{\mathcal{A}}, V_{\mathcal{A}}) = \Phi(P_{\mathcal{B}}, V_{\mathcal{B}}) = \Phi(P_{\mathcal{C}}, V_{\mathcal{C}}) = \theta}$$

Thus the zeroth law implies the existence of a property called temperature which has the same value across systems that are at thermal equilibrium. This is the working principle behind all thermometers.

2.3 Thermometry

Some common thermometers are constant volume gas thermometers, resistance thermometers, magnetization thermometers, linear/area/volume expansion thermometers. All this thermometers have one thing in common – a thermodynamic property X that varies linearly with temperature, I can write this as $X(\theta)$ but it is more useful to write the temperature in terms of this thermodynamic property as

$$\theta(X) = aX \tag{2.6}$$

where a is some constant.

Two fixed points are required to define a linear temperature scale. Given any two reference points (X_1, θ_1) and (X_2, θ_2), this is how you find the equation of that particular temperature scale using the information that the scale is linear (ie, its slope doesn't change)...

$$\boxed{\frac{\theta(X) - \theta_1}{X - X_1} = \frac{\theta_2 - \theta_1}{X_2 - X_1}} \tag{2.7}$$

Celsius scale is based on freezing and boiling point of water. But it was discarded from being a standard for the difficulty in reproducing the standard points as the temperature at which water freezes/boils changes with pressure.

Later, the Kelvin scale was introduced. For this scale one fixed point was the triple point of water (which happens at only one particular temperature and pressure) and the other fixed point is the theoretical minimum temperature a body could have. The latter was called 0 Kelvin and the former is 273.16 Kelvin.

Lets say I know the thermodynamic property of a material at the triple point temperature X_{TP}. I can then substitute $273.16K$ for $\theta(X_{\text{TP}})$ and obtain an expression for the constant a as

$$a = \frac{273.16\ K}{X_{\text{TP}}}$$

Substituting this in (2.6), I get

$$\theta(X) = \left(\frac{273.16\ [K]}{X_{\text{TP}}}\right) X \tag{2.8}$$

2.4 Equations of State

A system in thermal equilibrium do not have all independent variable and one arrives at a condition like the one obtained before —$\Phi(P, V) = \theta$. Some other common equations of state are

1. **Ideal Gas Law :**

$$PV - nRT = 0$$

2. **Van-der Waal's Equation :**

$$\left(P + a\frac{n^2}{V^2}\right)(V - nb) - nRT = 0$$

   where a is related to the molecular interactions and b is related to the finite size of the particles both of which the ideal gas law ignored.

3. **Virial Equation:**

$$P - \frac{Nk_BT}{V}\left[1 + \frac{N}{V}B_2(T) + \frac{N^2}{V^2}B_3(T) + ...\right] = 0$$

   where $B_2(T), B_3(T), ...$ are called virial coefficents.

4. **Curie's Equation of state of paramagnet**

$$M - \frac{CH}{T} = 0$$

   $M \longrightarrow$ Magnetization

   $H \longrightarrow$ Applied external magnetic field

   $C \longrightarrow$ Curie constant

   $T \longrightarrow$ Temperature

Given an equation of state, it can be solved for one variable in terms of the other variables like say volume expressed in terms of pressure and temperature–$V(P,T)$. Thermodynamic equilibrium ensures that infinitesimal changes in state variables doesn't change the equilibrium.

By theorem of partial differentials, if z is differentiable in x and y, it can be written as

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy$$

If it can't be written in this form, it is called a non-exact differential equation.

P, V and T represent actual variables of state in thermodynamic equilibrium and their differentials are exact differentials.

Writing $V(P,T)$

$$dV = \left(\frac{\partial V}{\partial T}\right)_P dT + \left(\frac{\partial V}{\partial P}\right)_T dP \tag{2.9}$$

The term $\left(\frac{\partial V}{\partial T}\right)_p$ is simply the volume thermal expansion and I can express it as

$$\begin{aligned} dV &= \beta V dT \\ \implies \frac{\partial V}{\partial T} &= \beta V \end{aligned}$$

Similarly, $\left(\frac{\partial V}{\partial P}\right)_T$ is the negative of inverse of bulk modulus (or compressibility – κ), with a volume term also, to make it volumetric strain instead of dV.

$$\begin{aligned} \kappa &= -\frac{dV/V}{dP} \\ \implies \left(\frac{\partial V}{\partial P}\right)_T &= -\kappa V \end{aligned}$$

Thus, I can write (2.9) as

$$dV = \beta V dT - \kappa V dP$$

Similarly, I can write Pressure and Temperature as

$$dP = \left(\frac{\partial P}{\partial T}\right)_V dT + \left(\frac{\partial P}{\partial V}\right)_T dV \tag{2.10}$$

$$dT = \left(\frac{\partial T}{\partial P}\right)_V dP + \left(\frac{\partial T}{\partial V}\right)_P dV \tag{2.11}$$

Theorems on Partial Differential Equations

The following equations are built on variables x,y and z that have a fundamental relationship of the kind $f(x, y, z) = 0$

1. **Reciprocal Theorem**

$$\left(\frac{\partial y}{\partial x}\right)_z = \frac{1}{\left(\frac{\partial x}{\partial y}\right)_z} \tag{2.12}$$

2. **Cyclic theorem on triple scalar product**

$$\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1 \tag{2.13}$$

This might seem counter-intuitive at first, but here's the proof

proof: Let there exist x, y and z such that $\exists\ f(x, y, z)$. I can write them in exact differential form as follows

$$dz = \left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy \tag{2.14}$$

Similarly, I can write dx as

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y dz$$

Substituing (2.14) in above, we get

$$dx = \left(\frac{\partial x}{\partial y}\right)_z dy + \left(\frac{\partial x}{\partial z}\right)_y \left[\left(\frac{\partial z}{\partial x}\right)_y dx + \left(\frac{\partial z}{\partial y}\right)_x dy\right]$$

$$\implies dx = \left(\frac{\partial x}{\partial y}\right)_z dy + dx + \left(\frac{\partial x}{\partial z}\right)_y \left(\frac{\partial z}{\partial y}\right)_x dy$$

$$\implies \boxed{\left(\frac{\partial x}{\partial y}\right)_z \left(\frac{\partial y}{\partial z}\right)_x \left(\frac{\partial z}{\partial x}\right)_y = -1}$$

3. **Test for Exactness:** Suppose I have a differential equation of the form

$$df = M(x, y)dx + N(x, y)dy \tag{2.15}$$

It is exact iff

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

proof: Since (2.15) is exact, I can compare it to (2.14) and write

$$M = \left(\frac{\partial f}{\partial x}\right)_y \quad \text{and} \quad N = \left(\frac{\partial f}{\partial y}\right)_x$$

Then, taking a partial derivative of M with respect of y and N with respect of x, we get

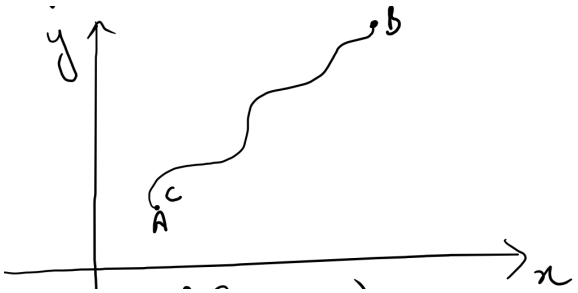
$$\frac{\partial M}{\partial y} = \frac{\partial^2 f}{\partial y \partial x} \quad \text{and} \quad \frac{\partial N}{\partial x} = \frac{\partial^2 f}{\partial x \partial y}$$

If a function is smooth and continuous, the second derivative would communte

$$\implies \frac{\partial^2 f}{\partial y \partial x} = \frac{\partial^2 f}{\partial x \partial y}$$
$$\implies \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

4. **Path independence:** An exact differential is path independent.

   proof: Say a system goes from state-A to state-B through some path-C. we get



$$\int_A^B df(x,y) = \int_A^B \left(\frac{\partial f(x,y)}{\partial x}\right)_y dx + \int_A^B \left(\frac{\partial f(x,y)}{\partial y}\right)_x dy$$

parameterizing x and y in terms of a single variable t, since $dx = \frac{dx}{dt}dt$ and $dy =$

$\frac{dy}{dt}dt$, we get...

$$\begin{aligned}\int_{t_A}^{t_B} df(x,y) &= \int_{t_A}^{t_B} \left(\frac{\partial f}{\partial x}\right)\frac{dx}{dt}\,dt + \int_{t_A}^{t_B}\left(\frac{\partial f}{\partial y}\right)\frac{dy}{dt}\,dt \\ &= \int_{t_A}^{t_B}\left(\frac{\partial f}{\partial x}\frac{dx}{dt} + \frac{\partial f}{\partial y}\frac{dy}{dt}\right)\,dt \\ &= \int_{t_A}^{t_B}\left(\frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy\right) \\ &= \int_{t_A}^{t_B} df\left(x(t), y(t)\right) = f(t_B) - f(t_A)\end{aligned}$$

This means that the value of integral only depends on the initial and final points and not the path taken between the two states ie, path independence.

3 The First Law of Thermodynamics

3.1 Internal Energy

Refers to the sum total of all energies of all particles that make up the system. It includes kinetic energy, vibrational energy, intermolecular potential energy, etc. For an ideal gas, we ignore the intermolecular attraction between the particles of the system and also for polyatomic species, we assume there is no vibration of bonds. In such a scenario, the Internal energy is given by the sum of the kinetic energies of the particles and therefore is only dependent on the temperature of the system.

$$U = \left(\frac{f}{2}\right)nRT \tag{3.1}$$

where f is the degrees of freedom for a molecule. At low temperatures where molecules don't vibrate,

Monoatomic : $f = 3$ (3 translational modes)
Diatomic : $f = 5$ (3 translational modes + 2 rotational modes)
Polyatomic : $f = 6$ (3 translational modes + 3 rotational modes)

The First Law

Finding the absolute value of the Internal Energy of a system is not merely as useful as finding the change in Internal Energy during a process. According to the first law of thermodynamics, there are only two ways of changing the internal energy of a system. They are by transfer of heat and by doing Work. That is

$$\boxed{\Delta U = Q + W_{\text{ext}}}$$

3.2 Doing Work on a gas

When we say that we are doing work on a gas, it implies that we are exerting external force (more precisely pressure) on a gas to decrease the volume occupied by the gas system. Similarly, when the gas does work, it pushes against an external force to expand its volume.

$$dW_{\text{ext}} = \vec{F} \cdot \vec{ds}$$
$$dW_{\text{ext}} = (P_{\text{ext}}\ \vec{A}\ ) \cdot \vec{ds}$$

During a compression or expansion of a volume, the surface that encapsulates that volume moves inwards or outwards therefore, the angle between Area element and displacement element is $0°$. Hence, I can write

$$dW_{\text{ext}} = P_{\text{ext}}\ (A \cdot ds)$$
$$dW_{\text{ext}} = P_{\text{ext}} \cdot dV$$

The derivation only yields us the expression for work done, it is upto to us to make the sign convention consistent with the first law. What I want is W_{ext} to be positive when an external force compresses the system and to be negative when the gas does work against an external force to expand its volume. Therefore, if V denotes the volume of the the gas, I want $W_{\text{ext}} > 0$ when $dV < 0$. After correction, I get

$$\boxed{dW = -P_{\text{ext}}\ dV}$$
$$W = -\int_{V_1}^{V_2} P_{\text{ext}}\ dV$$

Graphically, W_{ext} is given by the area under P-V graph.

Reversible and Irreversible Work

There are mainly two types of works - Reversible Work and Irreversible Work. Compression or expansion that involves a finite number of steps do Irreversible work on the system. This is because, if we compress a gas irreversibly, then let it expand, it won't return to the starting state of (P, V, T).

Meanwhile, reversible work involves doing a process infinitely slowly so that at the end of each step, the state of the gas is only infinitesimally different than before and the gas can return to its starting state if I allow it to. Also, during a reversible process, the gas it in equilibrium with the surrounding at every instant.

$$\implies P_{\text{int}} = P_{\text{ext}}$$

It can be seen from the P-V graph that, for the same change in volume, work done on the gas is minimum for reversible compression and maximum for a single step irreversible compression. Similarly, work done by the gas is maximum for reversible expansion and minimum for a single step irreversible expansion.

4 Heat Capacity

All objects don't experience the same increase in temperature for the same amount of heat added. Heat capacity(C) is defined as the heat required for a unit increase in the temperature of a system. That is...

$$C = \frac{dQ}{dT}$$

Heat added to increase the temperature of a body at constant pressure is called C_P and that at constant volume is called C_V.

$$C_P = \left(\frac{\partial Q}{\partial T}\right)_P \quad \text{and} \quad C_V = \left(\frac{\partial Q}{\partial T}\right)_V \tag{4.1}$$

According to the first law, $dU = dQ + dW = dQ - P_{\text{ext}}dV$. This implies that for an isochoric process, $dU = dQ$. Then...

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \tag{4.2}$$

Next, to relate C_P and C_V, I first start with internal energy. Since Internal energy is a state function and any two variables among (P, V, T) will fix the state of a system, I assume $U(V, T)$. I can write it as an exact differential as

$$dU = \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV$$

I substitute this for $dQ = dU + PdV$ to get..

$$\begin{aligned} dQ &= \left(\frac{\partial U}{\partial T}\right)_V dT + \left(\frac{\partial U}{\partial V}\right)_T dV + PdV \\ \implies dQ &= C_V dT + \left[\left(\frac{\partial U}{\partial V}\right)_T + P\right] dV \end{aligned}$$

Since internal energy only depends on temperature for an ideal gas, I can write $(\partial U/\partial V)_T = 0$. Thus,

$$dQ = C_V dT + PdV \tag{4.3}$$

To get C_V, I divide both sides with dT while keeping pressure constant

$$\begin{aligned} \left(\frac{dQ}{dT}\right)_P &= C_V + P\left(\frac{dV}{dT}\right)_P \\ \implies C_P &= C_V + P\left(\frac{nR}{P}\right) \end{aligned}$$

And thus, I arrive at

$$\boxed{C_P - C_V = nR} \tag{4.4}$$

Equation (4.3) writes dQ in terms of C_V. I can replace C_V with C_P in that equation to get in terms of C_P.

$$\begin{aligned} dQ &= (C_P - nR)dT + PdV \\ &= C_P dT - nRdT + PdV \end{aligned}$$

The ideal gas equation $PV = nRT$ when written in differential form yields $PdV + VdP = nRdT$. Thus...

$$dQ = C_P\ dT - VdP \tag{4.5}$$

5 Thermodynamic Processes

5.1 Isothermal Process

In an Isothermal process, Temperature of the system remains constant. That is, $\Delta T = 0$. Now since, the Internal energy U of a system is only function of the temperature, I can write

$$\Delta U_{\text{isothermal}} = 0$$

According to Ideal Gas law, $PV = nRT \implies PV = \text{constant}$, for Isothermal process.

To obtain expression for Work done on the gas during reversible isothermal process, I use

$$dW = -P_{\text{ext}}\ dV$$

$$W = -\int_{V_1}^{V_2} P_{\text{ext}}\ dV$$

For reversible processes, $P_{\text{ext}} = P_{\text{int}}$. And I can replace P_{int} with an expression in V using ideal gas law.

$$W = -\int_{V_1}^{V_2} \frac{nRT}{V}\ dV$$

$$W = -nRT\int_{V_1}^{V_2} \frac{1}{V}\ dV$$

$$\boxed{W_{\text{isothermal}} = nRT\ln\left(\frac{V_1}{V_2}\right)}$$

Now, since $\Delta U = 0 \implies W_{\text{ext}} = -Q$. Thus, the above equations holds for finding heat transferred as well.

Suppose the isothermal compressibility κ of substance is known to us and we have to derive an expression for work done when a certain amount of pressure is applied to our body. We know that

$$\kappa = \frac{-1}{V}\left(\frac{\partial V}{\partial P}\right)_T$$
$$\implies dV = -\kappa V \, dP$$

Substituting this for dV in the equation for work $dW = -P_{\text{ext}} dV$

$$\int dW = \int_{P_1}^{P_2} \kappa V P_{\text{ext}} \, dP$$

We treat V to be constant, because a finite change in pressure can be produced by very small values of volume change. Thus,

$$W = \frac{\kappa V}{2}(P_2^2 - P_1^2)$$

5.2 Adiabatic Process

In an adiabatic process, there is no heat exchange between the system and the surrounding. That is, $dQ = 0$. According to first law of thermodynamics,

$$dU = dQ + dW$$

substituting $dQ = 0$, we get

$$dU = \; dW_{\text{adia}} = -P_{\text{ext}} dV \tag{5.1}$$

There is an infinite number of adiabatic processes (need not be reversible) that take a system from one equilibrium state to another. But all this processes will have the same work since ΔU must be the same for all paths.

For a reversible Isochoric process, $dU = dQ_v = n \; c_v dT$. Now when I observe the equation $dU = n \, c_v dT$, I see that all terms are state functions, therefore although it has been derived for a specific kind of process (Isochoric), it is applicable to all reversible processes. Substituting that into dU in above equation, I get,

$$n \; c_v dT = -P_{\text{ext}} dV$$

Here, although I've used P_{ext}, for every quasi-static reversible process, $P_{\text{ext}} = P_{\text{int}}$. Replacing it in the above equation and using ideal gas law to write P_{int} in terms of V, T and n.

$$\begin{aligned} n\ c_v dT &= -\frac{nRT}{V} dV \\ \frac{c_v}{RT} dT &= -\frac{1}{V} dV \\ \int_{T_1}^{T_2} \frac{c_v}{RT} dT &= -\int_{V_1}^{V_2} \frac{1}{V} dV \end{aligned}$$

I assume that c_v is constant throughout the temperature range

$$\begin{aligned} \frac{c_v}{R} \int_{T_1}^{T_2} \frac{1}{T}\ dT &= \int_{V_2}^{V_1} \frac{1}{V}\ dV \\ \frac{c_v}{R}\ \ln\left(\frac{T_2}{T_1}\right) &= \ln\left(\frac{V_1}{V_2}\right) \\ \left(\frac{T_2}{T_1}\right)^{\frac{c_v}{R}} &= \left(\frac{V_1}{V_2}\right) \end{aligned}$$

Using the Ideal gas equation to simplify, I get...

$$\begin{aligned} \left(\frac{P_2 V_2}{P_1 V_1}\right)^{\frac{c_v}{R}} &= \left(\frac{V_1}{V_2}\right) \\ \left(\frac{P_2}{P_1}\right)^{c_v} &= \left(\frac{V_1}{V_2}\right)^{c_v + R} \\ \left(\frac{P_2}{P_1}\right) &= \left(\frac{V_1}{V_2}\right)^{\frac{c_p}{c_v}} \\ P_1 V_1^{\gamma} &= P_2 V_2^{\gamma} \end{aligned}$$

Therefore, for a **reversible** adiabatic process, $PV^{\gamma} = \text{constant}$. Where, $\gamma = \frac{c_p}{c_v}$. It is also worth noting that $\gamma = 1 + \frac{2}{f}$ where f is the degrees of freedom for the molecule.

Adiabatic Work

Work done on the gas during a reversible (quasi-static) adiabatic process is path independent and thus, the internal force driving the work can be assumed to be conservative. In that case, it can be written as the gradient of Potential energy, ie. $\vec{F}_{\text{gas}} = -\nabla U(r)$. Since

for a quasi-static process, $\vec{F}_{\text{gas}} = -\vec{F}_{\text{ext}}$, I can write

$$\begin{aligned}
\vec{F}_{\text{ext}} &= \nabla U(r) \\
\implies \vec{F}_{\text{ext}} &= \frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k} \\
\implies \vec{F}_{\text{ext}} \cdot d\vec{r} &= \left(\frac{\partial U}{\partial x}\hat{i} + \frac{\partial U}{\partial y}\hat{j} + \frac{\partial U}{\partial z}\hat{k}\right)(dx\ \hat{i} + dy\ \hat{j} + dz\ \hat{k}) \\
\implies dW_{\text{ext,adia}} &= \left(\frac{\partial U}{\partial x}dx + \frac{\partial U}{\partial y}dy + \frac{\partial U}{\partial z}dz\right) = dU
\end{aligned} \tag{5.2}$$

This can be thought of as a proof for (5.1) without the first law.

Remember that

$$dW = -P_{\text{ext}}\ dV = -P_{\text{int}}\ dV$$

for an adiabatic process $P_{int}\ V^{\gamma} =$ constant (say k)s

$$W = \int_{V_i}^{V_f} \frac{k}{V^{\gamma}}\ dV$$

$$W = k\left[\frac{V^{-\gamma+1}}{-\gamma+1}\right]_{V_i}^{V_f}$$

$$W = \left[\frac{k\ V_f^{-\gamma+1}}{-\gamma+1} - \frac{k\ V_i^{-\gamma+1}}{-\gamma+1}\right]$$

substitute $k = P_i\ V_i^{\gamma} = P_f\ V_f^{\gamma}$

$$W = \left[\frac{(P_f\ V_f^{\gamma})\ V_f^{-\gamma+1}}{-\gamma+1} - \frac{(P_i\ V_i^{\gamma})\ V_i^{-\gamma+1}}{-\gamma+1}\right]$$

$$W = \left[\frac{(P_f\ V_f^{\gamma})\ V_f^{-\gamma+1} - (P_i\ V_i^{\gamma})\ V_i^{-\gamma+1}}{1-\gamma}\right]$$

$$W = \frac{P_fV_f - P_iV_i}{1-\gamma} = \frac{nRT_f - nRT_i}{1-\gamma}$$

Therefore, work done on the gas during an adiabatic process is given by

$$\boxed{W_{\text{adiabatic}} = \frac{P_1V_1 - P_2V_2}{\gamma - 1} = \frac{nR(T_1 - T_2)}{\gamma - 1}}$$

6 The Carnot Cycle, Engine and Refrigerator

The Carnot engine is a reversible engine that operates between two fixed temperature reservoirs such that heat exchange happens only at these two fixed points. So first of all, I need two isothermal lines which are in contact with the reservoirs where heat exchange happens. Isothermal because, otherwise heat would be exchanged in a range of temperatures. Secondly, I need adiabatic lines that connect the two isotherms. Adiabatic so that heat exchange doesn't happen during that time. Also all these processes are carried out quasi-statically so that it is reversible at any instant.

Hence, the Carnot engine goes through four steps – Isothermal Expansion, Adiabatic Expansion, Isothermal Compression, Adiabatic Compression.

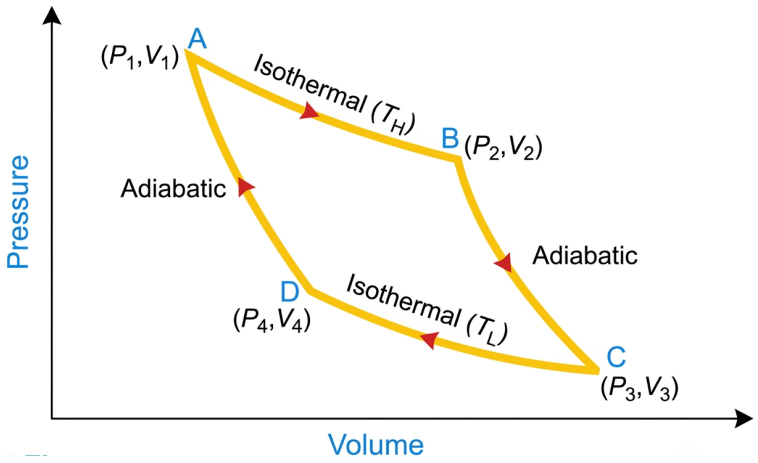


Figure 1: The Carnot Cycle

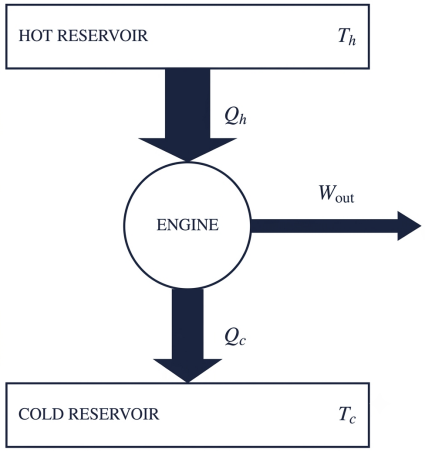


Figure 2: Schematic Diagram for every engine

Remember that during the isothermal arms, the temperature of our system is infinitesimally colder than the hot reservoir or hotter than the cold reservoir so as to let heat flow. If there was no temperature difference, heat wouldn't flow in and out of the reservoirs and if the temperature difference were to be finite, the process wouldn't be reversible.

6.1 Efficiency of a Carnot Engine

The efficiency of any engine is defined as the ratio of work done by the engine to the heat supplied by the hot reservoir.

$$\eta = \frac{|W_{\text{gas}}|}{|Q_H|} = \frac{|Q_H| - |Q_L|}{|Q_H|} = 1 - \frac{|Q_L|}{|Q_H|} \tag{6.1}$$

For a Carnot engine, I can further simplify this using our knowledge of isothermal and adiabatic processes as follows :

During the two isothermal arms, $\Delta U = 0$. I can then (using the first law), write Q_L and Q_H as

$$Q_H - W_{\text{gas},1\to2} = 0 \implies Q_H = W_{\text{gas},1\to2}$$

Since heat added to the system (Q_H) is taken as positive, $Q_H = |Q_H|$ in this case. Also, $W_{\text{gas},1\to2}$ is the work done *by the gas* during an isothermal process which I can simplify to

$$|Q_H| = nRT_H \ln\left(\frac{V_2}{V_1}\right) \tag{6.2}$$

Similarly,

$$Q_L - W_{\text{gas},3\to4} = 0 \implies Q_L = W_{\text{gas},3\to4}$$

In this case, Q_L is actually negative because our gas is losing heat to the cold reservoir during that step. Hence, $Q_L = -|Q_L|$. I again use the formula for isothermal work done by the gas to get

$$|Q_L| = -nRT_L \ln\left(\frac{V_4}{V_3}\right) = nRT_L \ln\left(\frac{V_3}{V_4}\right) \tag{6.3}$$

Since processes $2 \to 3$ and $4 \to 1$ are adiabatic, I can write the equation of state for adiabatic for these two arms as follows

$$\begin{aligned} P_2V_2^\gamma = P_3V_3^\gamma \quad &\text{and} \quad P_4V_4^\gamma = P_1V_1^\gamma \\ \implies T_HV_2^{\gamma-1} = T_LV_3^{\gamma-1} \quad &\text{and} \quad T_LV_4^{\gamma-1} = T_HV_1^{\gamma-1} \end{aligned}$$

Taking the ratio, I get

$$\frac{V_2}{V_1} = \frac{V_3}{V_4} \tag{6.4}$$

Taking the ratio of (6.2) and (6.3), and using our above result I get

$$\frac{|Q_L|}{|Q_H|} = \frac{T_L \ln(V_3/V4)}{T_H \ln(V_2/V_1)} = \frac{T_L}{T_H} \tag{6.5}$$

Substituting this in (6.1), I get the efficiency of a Carnot engine as

$$\boxed{\eta_{\text{Carnot}} = 1 - \frac{T_L}{T_H}} \tag{6.6}$$

This says that the efficiency of a Carnot engine is decided only by the temperatures between which it is operated on. All Carnot cycles operated between these two temperatures will have the same efficiency and no Carnot engine running between a finite temperature difference with $T_L \neq 0K$, can have 100% efficiency.

6.2 Carnot Refrigerator

Since the Carnot cycle was a reversible cycle, it can run as a refrigerator if run backwards. In that scenario, all arrows will be reversed – we will be doing work on the system as it takes in heat from the cold reservoir and gives it to the hot reservoir. The efficiency of such a refrigerator is same as the **reversible** heat pump counterpart.
However, a more useful term when it comes to refrigerators is COF or coefficient of performance defined as the heat supplied to the cold reservoir per unit work done on the system. It can take values from

$$COF = \frac{|Q_H|}{|W_{\text{ext}}|} = \frac{T_H}{T_H - T_L}$$

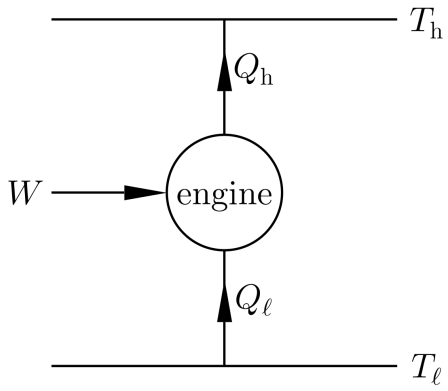


Figure 3: Carnot Refrigerator

6.3 Kelvin and Clausius Statements

Kelvin Statement: *"No cycle has heat absorbed from the hot reservoir and work delivered as its only effects."*

Clausius Statements: *"No cycle has heat moved from a cold body to a hot body as its only effects."*

The two statements are equivalent to each other. The only difference is that the former focuses on heat engines while the latter focuses on refrigerators.

It can be shown how if we violate one statement, the other also gets violated. For instance, Assume we have Kelvin violating engine that is running a refrigerator as shown below. Both engines taken together as one engine has a net effect of taking heat from a cold reservoir and pumping it to a hot reservoir without any external work and this violates the Clausius statement.

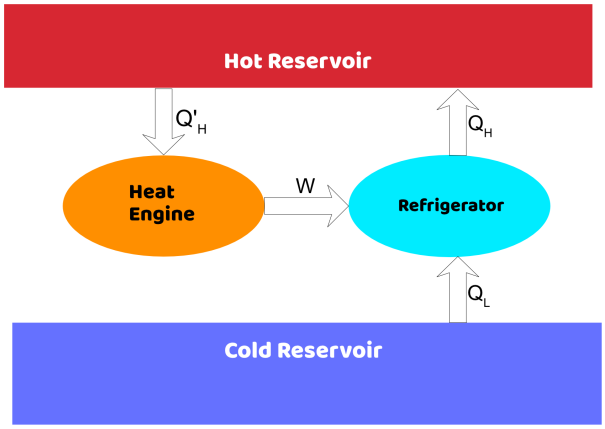


Figure 4: A Kelvin-statement violating heat engine running a refrigerator

6.4 Carnot Theorem

"No engine operating between two fixed temperatures has a better efficiency than a Carnot Engine operating between the two fixed temperatures"

Proof: Assume we have a Non-Carnot (irreversible) engine that is operating between two fixed temperatures and is driving a Carnot refrigerator as shown below.

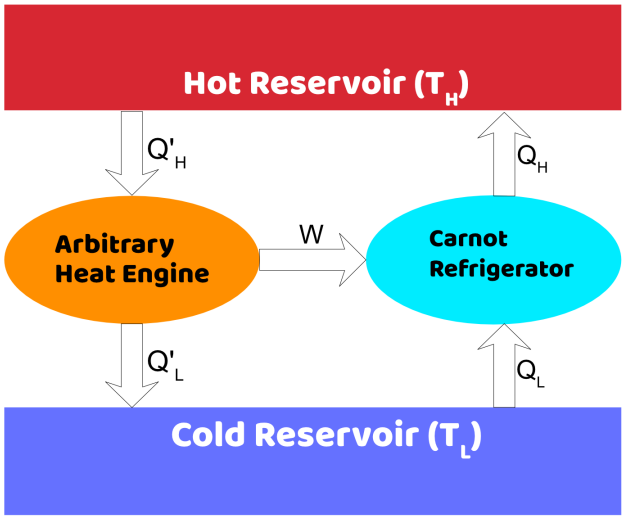


Figure 5: An irreversible engine used to drive a Carnot refrigerator

Assume that efficiency of the first engine (η') is more than the Carnot efficiency (η_{carnot})

$$\begin{aligned} & \eta' > \eta_{\text{carnot}} \\ \implies & \frac{|W|}{|Q'_H|} > \frac{|W|}{|Q_H|} \\ \implies & |Q'_H| < |Q_H| \end{aligned}$$

Hence, I get $|Q_H| - |Q'_H| > 0$. Similarly, I also arrive at $|Q_L| - |Q'_L| > 0$. If we take both engines together as our system, our combined engine seems to take a net heat from the cold reservoir and supply it to the hot reservoir without any external work. This violates the Clausius statement. Hence by contradiction, $\eta' \leq \eta_{\text{carnot}}$ always.

Note that we can't do the reverse process ie, a Carnot heat Engine used to drive an irreversible refrigerator because the efficiency of an irreversible refrigerator is not the same as its heat engine counter-part. But say my second heat engine is also reversible (not necessarily following the Carnot cycle), then I can reverse it to a refrigerator of the same efficiency and in that case, I'll get $\eta_{\text{reversible}} \leq \eta_{\text{carnot}}$ and $\eta_{\text{reversible}} \geq \eta_{\text{carnot}}$. Which implies $\eta_{\text{reversible}} = \eta_{\text{carnot}}$. This brings me to my second Carnot theorem...

– *"Any reversible engine operating between two fixed temperates has the same efficiency as a Carnot Engine."*

7 Entropy & The second law

7.1 General definition

Entropy(S) is a state function that gives the measure of disorder in a system. A highly ordered system has lower entropy than a highly disordered one. Speaking from a microscopic point of view, it is determined by the number of Microstates that a system can have.

Macrostate : Refers to the properties of the system that can be measured from a macroscopic point of view, like temperature, pressure, volume etc.

Microstate : Refers to the properties of the particles making up the system. Like exact position (x, y and z) and exact momentum (p_x, p_y and p_z) of each particle. That is, each particles microstate can be uniquely described by a single point on a 6-D cartesian system with the axes - x, y, z, p_x, p_y and p_z

7.2 Statistical Definition

Statistically, Entropy can be defined using Boltzmann's definition

$$S = k_B \ \ln(W)$$

Where, W is the number of available Microstates.

7.3 Change in Entropy

Although getting the exact value of entropy at any given moment requires the full Microstate description of a system, change in entropy can be found using macroscopic parameters also. Mathematically, change in entropy for **any reversible** process is given by

$$dS = \frac{dQ_{\text{rev}}}{T}$$

Below given list contains the expression for calculating change in entropy for some simple processes. All these can be derived from above equation.

Type of process	Change in Entropy
Cyclic Process	$\Delta S = 0$
Reversible Adiabatic	$\Delta S = 0$
Reversible Phase Phange	$\Delta S = \frac{\Delta H}{T}$
Reversible Isobaric Change	$\Delta S = n\, c_p \ln\left(\frac{T_f}{T_i}\right)$

It might seem counter-intuitive for the change in entropy for reversible adiabatic process to always be 0 because, if I let a gas undergo adiabatic expansion, the number of possible positional micro-states that the gas particles can occupy increases (because of the increased volume). It seems like Entropy should increase. But the catch here is that when the gas undergoes such an expansion, it cools down and therefore the range of available velocity(and momenta) shrinks. This reduces the number of available momentum Microstates. Both these effects perfectly cancel out leading to a net 0 change in entropy.

7.4 Clausius Inequality

Imagine a system that goes from a state (P_1, V_1, T_1) to a state (P_2, V_2, T_2) through two pathways - one reversible and one irreversible. Although paths are different, the change in state functions - ΔU, ΔS etc, are the same for both pathways.

$$dU = dQ_{\text{rev}} - P_{\text{int}}\ dV \tag{7.1}$$

$$dU = dQ_{\text{irrev}} - P_{\text{ext}}\ dV \tag{7.2}$$

subtracting both equations, I get...

$$dQ_{\text{rev}} - dQ_{\text{irrev}} = (P_{\text{int}} - P_{\text{ext}})\ dV$$

Now, if I observe the RHS of the equation, it must always be greater than zero. Because if $P_{\text{int}} > P_{\text{ext}}$, expansion happens. ie, $dV > 0$. $\implies$ product is positive. And also if $P_{\text{int}} < P_{\text{ext}}$, compression happens. ie, $dV < 0$ $\implies$ product is again positive.

$$\begin{aligned} (P_{\text{int}} - P_{\text{ext}})\ dV &> 0 \\ \implies dQ_{\text{rev}} - dQ_{\text{irrev}} &> 0 \\ dQ_{\text{rev}} &> dQ_{\text{irrev}} \end{aligned}$$

Now, since change in entropy of the system during this change is given by $dS = \frac{dQ_{\text{rev}}}{T_{\text{rev}}}$, I can replace dQ_{rev} in terms of dS.

For a reversible process, $T_{\text{sys}} = T_{\text{surr}}$ since a non-adiabatic system is in thermal equilibrium with the surrounding at all instants. Note that this entropy change refers to the change in entropy as the system goes from the initial state to final state and is applicable to both the reversible and irreversible paths.

$$\begin{aligned} T_{\text{surr}}\ dS &> dQ_{\text{irrev}} \\ dS &> \frac{dQ_{\text{irrev}}}{T_{\text{surr}}} \end{aligned}$$

Now, I integrate both sides over an entire cycle. ie, take the closed loop integral.

$$\oint dS > \oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}}$$

Since, entropy is a state function, its integral over a complete loop must be zero. That is,

$$\boxed{\oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}} < 0}$$

This is the Clausius Inequality. An important thing to note here is why the temperature used here is T_{surr} and not T_{sys}

Since the Surrounding acts as a temperature reservoir and are internally reversible, any process that happens to the Surrounding can be considered reversible. Therefore, the expression for change in entropy for the surrounding is not only for heat added in reversible processes but all processes.

$$\Delta S_{\text{surr}} = \frac{Q}{T_{\text{surr}}}$$

In this case, $dQ_{\text{surr}} = dQ_{\text{irrev}}$ of the system. Therefore, during one full cycle of the system, the change in entropy of the surrounding is given by

$$\Delta S_{\text{surr}} = \oint \frac{dQ_{\text{surr}}}{T_{\text{surr}}} = \oint \frac{-dQ_{\text{irrev}}}{T_{\text{surr}}}$$

7.5 The Second Law of Thermodynamics

It is clear from above that ΔS_{surr} can be written as

$$\Delta S_{\text{surr}} = -\oint \frac{dQ_{\text{irrev}}}{T_{\text{surr}}}$$

and according to Clausius Inequality, this expression has to be greater than 0.

If I now introduce another term $\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$. I see that for **all reversible processes,** $\Delta S_{\text{univ}} = 0$ and for **all irreversible processes,** $\Delta S_{\text{univ}} > 0$

To understand why this works, lets think of a common scenario, I have a hot closed cup of tea at $T_{\text{cup}} = 60°\text{C}$ exposed to the surrounding at $T_{\text{surr}} = 20°\text{C}$. What happens now is pretty obvious, the tea loses heat through the cup walls until it reaches thermal equilibrium with the surrounding. Let Q be the magnitude of heat that is transferred.

$$\Delta S_{\text{sys}} = -\frac{Q}{T_{\text{cup}}}$$

$$\Delta S_{\text{surr}} = \frac{Q}{T_{\text{surr}}}$$

$$\Delta S_{\text{univ}} = \Delta S_{\text{sys}} + \Delta S_{\text{surr}}$$

$$\Delta S_{\text{univ}} = -\frac{Q}{T_{\text{cup}}} + \frac{Q}{T_{\text{surr}}}$$

$$\Delta S_{\text{univ}} = Q\left(\frac{1}{T_{\text{surr}}} - \frac{1}{T_{\text{cup}}}\right)$$

since $T_{\text{cup}} > T_{\text{surr}}$,

$$\left(\frac{1}{T_{\text{surr}}} - \frac{1}{T_{\text{cup}}}\right) > 0$$

$$\Delta S_{\text{univ}} > 0$$

This also explains why heat wouldn't flow from a colder body to a hotter body. Although it is allowed by the First Law, It would imply that $\Delta S_{\text{univ}} < 0$. That would violate the Second Law.

8 Thermodynamic Functions

Thermodynamic functions are the different state variables we can define for a thermodynamic system in equilibrium. Since they are state variables, I can write them as exact differentials. Internal Energy(U) is an example of such a thermodynamic function. We get the other functions by something called Legendre Transformations

Legendre Transforms

Given a function $f(x, y)$ such that

$$df = \left(\frac{\partial f}{\partial x}\right)_y dx + \left(\frac{\partial f}{\partial y}\right)_x dy = Pdx + Qdy$$

P and x are one pair of conjugate variables and Q and y are another pair of conjugate variables. Legendre Transforms let you interchange the variables of a single conjugate pair.

Say I want to interchange P and x. Then, I take first take the combined differential $d(P \cdot x)$. I get

$$d(P \cdot x) = Pdx + xdP$$

If I now subtract this from df, I get

$$df - d(Px) = Pdx + Qdy - (Pdx + xdP) = Pdx - xdP$$

If I now write $df - d(Px) = d(f - Px) = dg$, I can write

$$dg = Pdx - xdP$$

Another exact differentiable function given by $g = f - Px$. This is then another thermodynamic function.

8.1 Enthalpy

Starting from the exact differential equation for internal energy(U),

$$dU = dQ - PdV$$

For my first thermodynamic function, I have to replace P and V in the above equation.

So I take $d(PV) = PdV + VdP$. If I now add this to dU, I get...

$$\begin{aligned} dU + d(PV) &= dQ - PdV + PdV + VdP \\ d(U + PV) &= dQ + VdP \\ dH &= dQ + VdP \end{aligned} \tag{8.1}$$

Where $\boxed{H = U + PV}$ is called the enthalpy of our system. However, just like internal energy, knowing its change during a process is more important than its raw value at any instant.

Physical Meaning: Enthalpy is the heat content of the system. Physically it gives the total energy of a system to exist in this universe, to push against an external atmospheric pressure and claim its rightful volume.

When pressure is constant, enthalpy equals heat added. That is, $dH = dQ_p$. It can then be defined as the heat added at constant pressure.

Why enthalpy? Most reactions happen exposed to the atmosphere, even if the volume of reactants change during the reaction, the pressure it exerts outwards always match the atmospheric pressure. In this case, it is easier to deal with Enthalpy when talking about heat of reactions.

8.2 Helmholtz Free Energy

We once again start with the definition of Internal energy $dU = dQ - PdV$. If I use $dS = dQ/T$, I get

$$dU = TdS - PdV$$

This thermodynamic function is obtained by reversing S and T.

So I take $d(TS) = TdS + SdT$. If I now subtract this from dU, I get...

$$\begin{aligned} dU - d(TS) &= TdS - PdV - (TdS + SdT) \\ d(U - TS) &= -PdV - SdT \\ dF &= -PdV - SdT \end{aligned} \tag{8.2}$$

Where $\boxed{F = U - TS}$ is called the Helmholtz free energy of our system.

8.3 Gibbs Free Energy

Expanding dQ in (8.1) using TdS, I get

$$dH = TdS + VdP$$

Gibbs free energy is obtained by replacing T and S.

So I take $d(TS) = TdS + SdT$. If I now subtract this from dH, I get...

$$\begin{aligned} dH - d(TS) &= TdS + VdP - (TdS + SdT) \\ d(H - TS) &= VdP - SdT \\ dG &= VdP - SdT \end{aligned} \tag{8.3}$$

Where $\boxed{G = H - TS}$ is called the Gibbs free energy of our system.

Reactions that happens at constant temperature and pressure (like phase changes, fusion, diffusion, boiling) will have constant Gibbs free energy.